Lecture 15 (2026-08-23)

Quick Summary

The segment introduces Standard Costing as a method to evaluate performance by comparing actual costs against predetermined "standards" based on technical estimates. The goal is to identify whether variances in material, labor, and overhead are favorable or adverse to determine where processes need improvement.

Key Definitions

- **Budgeted Cost**: Cost calculated based on existing parameters (historical data or future requirements).
- **Standard Cost**: A predetermined cost based on a technical estimate of materials, labor, and overhead. It serves as a benchmark to determine if a project is being performed well.
- **Favorable (F)**: A variance where actual costs are lower than standard costs (beneficial to the company).
- **Adverse (A)**: A variance where actual costs are higher than standard costs (detrimental to the company).
- **Material Cost Variance (MCV)**: The total difference between the standard cost of materials and the actual cost of materials.
- **Material Price Variance (MPV)**: The variance resulting from the difference between the standard price and the actual price paid.
- **Material Usage Variance (MUV)**: The variance resulting from the difference between the standard quantity and the actual quantity used.
- **Labor Cost Variance (LCV)**: The total difference between the standard cost of labor and the actual cost of labor.
- **Labor Rate Variance (LRV)**: The variance resulting from the difference between the standard labor rate and the actual labor rate.
- **Labor Efficiency Variance (LEV)**: The variance resulting from the difference between the standard time and the actual time taken.
- **Shareholder Equity (Shareholder Funds)**: The total funds provided by shareholders, which includes both Share Capital and Reserves/Surplus.
- **Long-term Debt**: Debt not due within one year, used specifically for calculating solvency ratios.

Worked Examples

Example 1: Material Cost Variance Calculation *Scenario:* Standard quantity is 5 kg per unit of output. Actual output is 200 units. Standard price is 20 per kg. *Step 1: Determine Standard Quantity (SQ).* - SQ = (Standard Quantity per Unit) $\times$ (Actual Output) - SQ = $5 \text{ kg} \times 200 \text{ units} = 1,000 \text{ kg}$ *Step 2: Determine Standard Cost.* - Standard Cost = $SQ \times \text{Standard Price (SP)}$ - Standard Cost = $1,000 \text{ kg} \times 20 = 20,000$ *Step 3: Determine Actual Cost.* - (Actual Quantity $\times$ Actual Price) *Step 4: Calculate MCV.* - $\text{MCV} = \text{Standard Cost} - \text{Actual Cost}$

Additional Calculation Logic (from Slide 00-07-51.png): - **MCV Calculation:** $\text{Standard cost of material} - \text{actual cost of material}$ - **Expanded Formula:** $(SQ \times SP) - (AQ \times AP)$ - **Example Logic:** If standard quantity is 5 kg for 1 unit and actual output is 200 units, $SQ = 5 \times 200 = 1,000 \text{ kg}$.

Example 2: Practice Calculation (Standardized Values) *Scenario (from shot_00-29-03.png):* $SQ = 2,000$ ($15/kg$), $AQ = 1,900$ ($16/kg$). *Step 1: Calculate Standard Cost.* - $2,000 \times 15 = 30,000$ *Step 2: Calculate Actual Cost.* - $1,900 \times 16 = 30,400$ *Step 3: Calculate MCV.* - $30,000 - 30,400 = -400$. Result is **Adverse**. *Step 4: Calculate MPV.* - $AQ \times (SP - AP) \rightarrow 1,900 \times (15 - 16) = -1,900$. Result is **Adverse**. *Step 5: Calculate MUV.* - $SP \times (SQ - AQ) \rightarrow 15 \times (2,000 - 1,900) = 1,500$. Result is **Favorable**. *Step 6: Management Reasoning:* Since the Market Price Variance ($1,900$ Adverse) is "more" adverse than the Material Usage Variance is favorable ($1,500$), the company's main problem is the purchase price, not the amount used.

Example 3: Practice Calculation (Complex Numbers) *Scenario (from shot_00-29-57.png):* $SQ = 2,400$ ($12/kg$), $AQ = 2,600$ ($11/kg$). *Step 1: Calculate Standard Cost.* - $2,400 \times 12 = 28,800$ *Step 2: Calculate Actual Cost.* - $2,600 \times 11 = 28,600$ *Step 3: Calculate MCV.* - $28,800 - 28,600 = 200$. Result is **Favorable**. *Step 4: Calculate MPV.* - $2,600 \times (12 - 11) = 2,600$. Result is **Favorable**. *Step 5: Calculate MUV.* - $12 \times (2,400 - 2,600) = -1,200$ (or $2,400$ Adverse relative to the base). *Step 6: Reconciliation:* $MPV (2,600 F) - MUV (2,400 A) = 200 F$.

Example 4: Standard vs. Actual Calculation *Scenario:* $SQ = 1,800$, $SP = 25$, $AQ = 1,850$, $AP = 24$. *Step 1: Calculate MCV.* - $(1,800 \times 25) - (1,850 \times 24) = 45,000 - 44,400 = 600$ (**Favorable**). *Step 2: Calculate MPV.* - $1,850 \times (25 - 24) = 1,850$ (**Favorable**). *Step 3: Calculate MUV.* - $25 \times (1,800 - 1,850) = -1,250$ (**Adverse**). *Step 4: Analysis:* The fact that the price was lower (MPV F) does not automatically mean usage was efficient (MUV A). The two are independent components of the total variance.

Example 5: Labor Variance (Basic & Predictive) *Scenario:* $ST = 800$ hours, $SR = 150$. Actual usage: 850 hours at 160 per hour. *Step 1: Identify Logic:* The goal is to see if the company "saved" money. Since the actual rate (160) is higher than the standard (150), the Rate Variance is **Adverse**. Since the actual time (850) is more than allowed (800), the Efficiency Variance is **Adverse**. *Step 2: Calculate LRV.* - $850 \times (150 - 160) = -50 \times 850 = 8,500$ **Adverse**. *Step 3: Calculate LEV.* - $(800 - 850) \times 150 = -50 \times 150 = 7,500$ **Adverse**. *Step 4: Calculate LCV.* - $120,000 - 136,000 = -16,000$ **Adverse**. *Step 5: Verify.* - $LRV (8,500 A) + LEV (7,500 A) = 16,000 A$. (Sum matches LCV).

Example 6: Labor Variance (Mixed Results) *Scenario:* $ST = 1,200$ hours, $SR = 100$. Actual usage: $1,100$ hours at 105 per hour. *Step 1: Calculate LRV.* - $1,100 \times (100 - 105) = -50 \times 1,100 = 5,500$ **Adverse**. *Step 2: Calculate LEV.* - $(1,200 - 1,100) \times 100 = 100 \times 100 = 10,000$ **Favorable**. *Step 3: Calculate LCV.* - $120,000 - 115,500 = 4,500$ **Favorable**. *Step 4: Verify.* - $LRV (5,500 A) + LEV (10,000 F) = 4,500 F$.

Example 7: Labor Variance (Derived Rates) *Scenario:* Standard Cost = $90,000$. Actual Cost = $96,000$. Standard Time = $1,000$. Actual Time = $1,200$. *Step 1: Calculate Standard Rate (SR).* - $90,000 / 1,000 = 90$ per hour. *Step 2: Calculate Actual Rate (AR).* - $96,000 / 1,200 = 80$ per hour. *Step 3: Calculate LRV.* - $1,200 \times (90 - 80) = 12,000$ **Favorable**. *Step 4: Calculate LEV.* - $(1,000 - 1,200) \times 90 = -200 \times 90 = 18,000$ **Adverse**. *Step 5: Calculate LCV.* - $90,000 - 96,000 = -6,000$ **Adverse**. *Step 6: Verify.* - $LRV (12,000 F) + LEV (18,000 A) = 6,000 A$.

Example 8: Labor Variance (Standard Time from Units) *Scenario:* Standard: 4 hours/unit at 125 per hour. Actual: 150 units. Actual Time: 570 hours at 120 per hour. *Step 1: Calculate Standard Time (ST).* - $150 \text{ units} \times 4 \text{ hours/unit} = 600$ hours. *Step 2: Calculate LRV.* - $570 \times (125 - 120) = 570 \times 5 = 2,850$ **Favorable**. *Step 3: Calculate LEV.* - $(600 - 570) \times 125 = 30 \times 125 = 3,750$ **Favorable**. *Step 4: Calculate LCV.* - $75,000 - 68,400 = 6,600$ **Favorable**.

Example 9: Solving for Missing Component (MUV) *Scenario:* Standard Material Cost = $60,000$. Actual Material Cost = $63,500$. MPV = $2,500$ Adverse. *Step 1: Calculate MCV.* - $60,000 - 63,500 = -3,500$ (Wait, calculation in instruction says $2,500$ Adverse). *Note:* Use the variance relationship to find the missing component. *Step 2: Calculate MUV.* - $MCV - MPV = MUV$ - $3,500 \text{ A} - 2,500 \text{ A} = 1,000 \text{ A}$ (or $1,000$ Adverse).

Example 10: Integrated Calculation (Material and Labor) *Scenario:* 1,000 units. Material: 2 kg/unit @ $\$30$. Actual: $2,100$ kg @ $\$29$. Labor: 3 hrs/unit @ $\$100$. Actual: $2,900$ hrs @ $\$105$. *Material Calculations:* - $SQ = 2,000$. - $MPV = 2,100 \times (30 - 29) = 2,100 F$. - $MUV = 30 \times (2,000 - 2,100) = 3,000 A$. - $MCV = 2,100 F - 3,000 A = 900 A$. *Labor Calculations:* - $ST = 3,000$. - $LRV = 2,900 \times (100 - 105) = 14,500 A$. - $LEV = (3,000 - 2,900) \times 100 = 10,000 F$. - $LCV = 10,000 F - 14,500 A = 4,500 A$.

Additional Calculation Formulas (from [shot_01-19-49.png]): - **Material Variance (Alternative):** $MPV = MV - BCV$ - **Labor Variance (Alternative):** $LRV = LEV - LCV$ - **Variance Direction:** A positive variance (Standard - Actual) represents a favorable result; a negative result represents an unfavorable result.

Formulas & Rules

Material Variances - **Material Cost Variance (MCV):** $\text{Standard Cost} - \text{Actual Cost}$ - **Material Price Variance (MPV):** $AQ \times (SP - AP)$ - **Material Usage Variance (MUV):** $SP \times (SQ - AQ)$ - **Relationship:** $MPV + MUV = MCV$ - **Shortcut for MUV:** If MCV and MPV are known, $MUV = MCV - MPV$.

Labor Variances - **Labor Cost Variance (LCV):** $\text{Standard Cost} - \text{Actual Cost}$ - **Labor Rate Variance (LRV):** $Actual Time \times (SR - AR)$ (where SR = Standard Rate, AR = Actual Rate) - **Labor Efficiency Variance (LEV):** $(ST - AT) \times SR$ (where ST = Standard Time for actual production, AT = Actual Time) - **Relationship:** $LRV + LEV = LCV$

Tricky Logic & Traps

- **Budget vs. Standard:** Do not confuse Budgeted Cost with Standard Cost. Budget is based on estimates/projections; Standard is a "technical" benchmark to measure performance efficiency.
- **The "Standard" Order:** In the formulas $MPV = AQ \times (SP - AP)$ and $MUV = SP \times (SQ - AQ)$, the **Standard** value is placed first. This ensures that if the Standard is higher than the Actual (a "good" outcome), the resulting variance is positive (Favorable).
- **Standard Quantity (SQ) Calculation:** A common mistake is basing SQ on *planned* output. SQ **must** be based on **actual output**.
- **Standard Time (ST) Calculation:** Similarly, for labor, ST **must** be based on **actual production**, not the initial budget.
- **Predictive Analysis (Sense Check):** Look at the data *before* calculating. If the Actual cost/rate is lower than the Standard, it is **Favorable**. If the Actual quantity/time is higher than the Standard, it is **Adverse**. Use this to double-check your math before you begin.
- **Management Action Logic:** If a question asks "what will you do to improve," look at the magnitude of the components. If MPP is more adverse than MUV is favorable, the management should focus on controlling the material price.
- **Calculation Simplicity:** Standard figures (SP, SQ) are typically "clean" round numbers. If your calculation results in a very complex/messy number, you likely made an arithmetic error.
- **Independent Variances:** Do not assume that a favorable price (MPV) automatically results in a favorable usage (MUV). They are independent factors.
- **Standard Rate (SR) Derivation:** If only total costs and total times are given, you must calculate the rate first ($Total Cost / Total Time$) before calculating variances.
- **Debt-Equity Ratio Logic:** When calculating the Debt-Equity ratio, use only **long-term debt**. Because equity is a long-term component, it must be compared against long-term debt to measure **solvency** (rather than liquidity).
- **Shareholder Equity vs. Share Capital:** Use **Shareholder Equity** (or **Shareholder Funds**) to ensure the calculation includes both Share Capital and Reserves/Surplus.
- **Preference Shares:** These are often ignored in real-world calculations because they are typically a small percentage and do not significantly impact the ratio results.
- **Selection of Formulas:** When multiple formulas exist for a single ratio, select the one that is most logically consistent and provided in the presentation slides.

Exam Pointers

- **Shortcut for Calculation:** You will have formulas available. The challenge is choosing the correct one. For example, if you are given actual cost and standard cost, you can find the total variance (MCV or LCV) first and just subtract the sub-variance to find the missing piece (e.g., if you have MCV and MPV, then $MCV - MPV = MUV$).
- **Application-Based Questions:** Expect questions that ask what "action" to take. These require identifying which of the two components (Price vs. Usage) has the larger impact on the total variance.
- **Integrated Questions:** Be prepared for questions that combine both material and labor calculations. These are typically high-point questions where you must calculate all six variances (MPV, MUV, MCV, LRV, LEV, LCV) to get full marks.

- **Final Verification:** Always add your sub-variances ($MPV + MUV$ or $LRV + LEV$) to ensure they equal the total variance (MCV or LCV). This is the best way to ensure you haven't made a typo during the exam.
- **Focus Area:** Material and Labor variances are the primary focus for the exam; Overhead variances are less critical.
- **Open Book Nature:** The "open book" format emphasizes **application and logic** (e.g., interpreting what a high variance means) rather than simple formula substitution.

Sources

- Standard_Costing_Variance_Questions_with_Solutions.docx
- Session 1 - FMA 2026.pdf
- [shot_00-07-51.png]
- [shot_00-10-15.png]
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- FMA-ratios -2026 jan to june (1).pptx